4-DOF Robot Arm

SolidWorks · FDM printing · Arduino · SG90 servos · joystick control

PHOTO OF THE FINISHED ARM

A 3D-printed desktop robot arm with four degrees of freedom - base, shoulder, elbow and a gear-driven gripper - controlled in real time with two analog joysticks. Designed in SolidWorks, printed in PLA, wired and programmed on Arduino.

The main engineering constraint was the torque of the SG90 micro servo. Link lengths, gripper mass and the working payload were sized from a shoulder torque calculation rather than chosen by eye.

4 DOF
base, shoulder, elbow, gripper

~22 cm
reach, shoulder axis to jaws

~17 g
max payload at full reach

28 %
of SG90 stall, arm's own weight

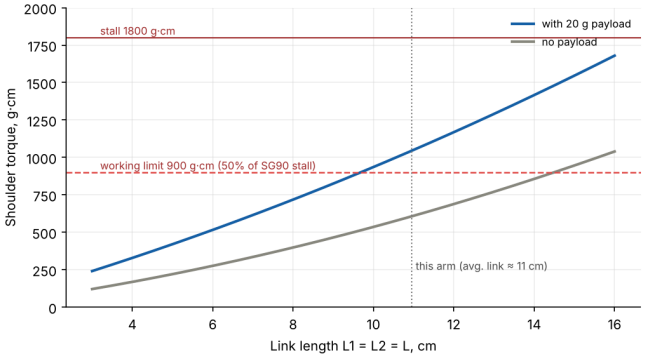
Shoulder torque calculation

Worst case: arm fully extended. Weighed part masses, lever arms from the SolidWorks assembly:

$$M = \Sigma\, m_i \cdot r_i$$

Element	m, g	r, cm	M, g·cm
Link 1	9.73	5.0	49
Elbow servo	9	10.5	95
Link 2	9.10	14.7	134
Gripper servo	9	17.6	158
Gripper fingers	3.79	19.7	75
Arm only			**510**
Payload	20	21.9	438
With payload			**948**

SG90 stall torque 1.8 kg·cm. Working limit taken as 50 % (900 g·cm) because stall means the motor is stopped and overheating, and clone servos are often weaker.



Finding: 20 g at full reach gives 948 g·cm, just over the working limit. Max payload within the limit:

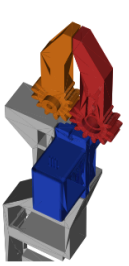
$$m_max = (900 - 510) / 21.9 \approx 17\ g$$

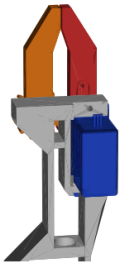
Fix: a 50 g counterweight 4 cm behind the shoulder brings the loaded case to 748 g·cm. 71 % of the torque comes from the gripper end.

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Gear-driven gripper

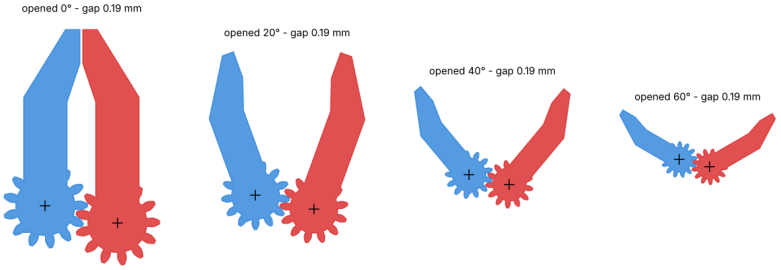
Two identical spur gears couple the fingers so both move symmetrically from one servo. One finger is mounted on the servo horn, the other rotates on an M2 axis in the same part.





Parameter	Value
Teeth z / module m	13 / 1.17 mm
Pressure angle	20°, involute
Centre distance a = m·z	15.41 mm
FDM clearance	+0.2 mm, 0.06 mm backlash

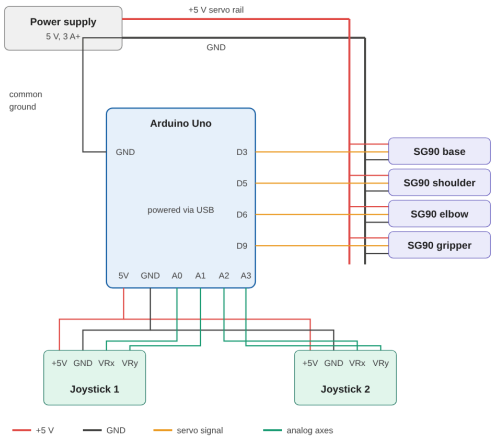
The module was derived from the centre distance fixed by the existing part, not chosen first. Mesh and collisions checked from 0° to 60° opening in 5° steps.



Design iterations

Version	Problem	Fix
Gripper v1	Fingers lost sync at large angles: the second gear sat on a thin pin with no rigid frame, gear forces pushed the axes apart and teeth skipped.	Both axes in one rigid part, full 360° gears, integrated spacer boss as bearing surface.
Power v1	Servos powered from the Arduino 5 V pin.	Separate 5 V supply with common ground.

Electronics and firmware



- **Velocity control:** stick deflection sets joint speed, so the arm holds its pose when released.
- **Auto-calibration** of joystick centres at start-up (16-sample average).
- **Dead zone** and **per-joint soft limits** keep servos away from mechanical stops.
- **Staggered servo attach** avoids a current spike at power-on.
- Wall thicknesses are multiples of the 0.4 mm extrusion width; horn pockets +0.15 mm for FDM tolerance, verified with a test print.

Gripper v1 fingers: Servo Mini Arm by Luca Dilo. Gripper v2 finger geometry generated parametrically (CadQuery, AI-assisted) from the assembly dimensions.

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